

***In silico* Analysis of Transcription Factor Repertoire and Prediction of Stress Responsive Transcription Factors in Soybean**

KEIICHI Mochida^{1,*}, TAKUHIRO Yoshida¹, TETSUYA Sakurai¹, KAZUKO Yamaguchi-Shinozaki², KAZUO Shinozaki¹, and LAM-SON PHAN Tran^{1,*}

RIKEN Plant Science Center, 1-7-22 Suehiro-cho, Tsurumi-ku, Yokohama, Kanagawa 230-0045, Japan¹ and Japan International Center of Agricultural Sciences, Ibaraki 305-8686, Japan²

(Received 27 July 2009; accepted 5 October 2009; published online 2 November 2009)

Abstract

Sequence-specific DNA-binding transcription factors (TFs) are often termed as ‘master regulators’ which bind to DNA and either activate or repress gene transcription. We have computationally analysed the soybean genome sequence data and constructed a proper set of TFs based on the Hidden Markov Model profiles of DNA-binding domain families. Within the soybean genome, we identified 4342 loci encoding 5035 TF models which grouped into 61 families. We constructed a database named SoybeanTFDB (<http://soybeantfdb.psc.riken.jp>) containing the full compilation of soybean TFs and significant information such as: functional motifs, full-length cDNAs, domain alignments, promoter regions, genomic organization and putative regulatory functions based on annotations of gene ontology (GO) inferred by comparative analysis with *Arabidopsis*. With particular interest in abiotic stress signalling, we analysed the promoter regions for all of the TF encoding genes as a means to identify abiotic stress responsive *cis*-elements as well as all types of *cis*-motifs provided by the PLACE database. SoybeanTFDB enables scientists to easily access *cis*-element and GO annotations to aid in the prediction of TF function and selection of TFs with functions of interest. This study provides a basic framework and an important user-friendly public information resource which enables analyses of transcriptional regulation in soybean.

Key words: soybean; transcription factors; abiotic stress; database

1. Introduction

Sequence-specific DNA-binding transcription factors (TFs) are the key molecular switches that control or influence many of the biological processes such as development, growth, cell division and responses to environmental stimuli in a cell or organism. By being capable of activating or repressing transcription of multiple target genes, they affect the metabolism, physiological balance and progression in cells and the responses of cells to the environment.^{1–3} TFs form

complex regulatory networks at the transcriptional level and through protein–protein interactions among themselves or with proteins of other classes. Protein–protein interactions may also form with other transcriptional regulators such as chromatin remodelling/modifying proteins to recruit or block access of RNA polymerases to the DNA template. The specific interactions between TFs and a family of *cis*-regulatory sequences described by a consensus motif play a central part in how genetic regulatory proteins affect spatial and temporal gene expression.⁴ Additionally, alterations in the activity and regulatory specificity of TFs are emerging as a major source of diversity and evolutionary adaptation.^{5,6}

In the past decade, the availability of complete genome sequences and the development of

Edited by Dr. Katsumi Isono

* To whom correspondence should be addressed. Tel. +81 45-503-9593. E-mail: mochida@psc.riken.jp (K.M.) or tran@psc.riken.jp (L.-S.P.T.)

high-throughput experimental techniques have enabled scientists to compile complementary information describing the function and organization of TF regulatory systems in a number of organisms. The identification, characterization and classification of TFs at the genome-wide level will provide an important resource for researchers who are interested in studying the regulation of gene expression. Similar to other proteins, TFs are comprised evolutionarily conserved units called 'domains', which belong to families that can occur in many different proteins. The majority of TFs can be grouped into a number of different families according to the specific type of DNA-binding domain (DBD) that is present within their sequence.^{7–10} Using bioinformatics approaches, computational studies have documented valuable TF repertoires by searching for genes containing DBDs within individual organisms ranging from prokaryotes to eukaryotes or by searching across all completely sequenced genomes.^{7,8,10–18}

In plants, ~7% of the genome encodes putative TFs.¹⁹ Despite their importance as a fundamental component of biological systems, the TF repertoires for many plant genomes remain largely unknown and understudied. Analyses of expressed sequence tag (EST) and genome sequence databases have indicated that legumes encode more than 2000 TFs per genome. At the present time, less than 1% of these putative TFs have been genetically and functionally characterized.¹⁹ Our basic knowledge of TFs and their role in transcriptional regulation is derived from molecular biological and genetic investigations. Proper characterization of particular TFs often requires a detailed study in the biological context of a whole TF family, since functional redundancy is a common occurrence within TF families.^{20–24} Furthermore, since TFs control the expression of the genome, it is not possible to completely understand their function without performing detailed functional studies at a genome-wide level.^{7,25–27}

Soybean (*Glycine max* L.) is a nutritionally important crop which provides an abundant source of oil and protein for worldwide human consumption.^{28–31} In addition, soybean is also viewed as an attractive crop for the production of renewable fuels such as biodiesel. Due to its symbiosis with nitrogen fixing bacteria, soybean can fix atmospheric nitrogen and therefore requires minimal input of nitrogen fertilizer. Agricultural dependence on nitrogen fertilizer often accounts for the single largest energy input in agronomic practices.³² With the recent completion of the soybean genomic sequence (<http://www.phytozome.net/soybean#C> Soybean Genome Project, DOE Joint Genome Institute), the identification, isolation and functional analysis of important genes will be accelerated. From a biotechnology

perspective, this resource will be especially important for studying regulatory genes involved in plant productivity, seed quality, nitrogen fixation and the sensing/response and adaptation to the environment. Within the soy genome model, ~975 Mb has been captured in 20 chromosomes and 66 153 protein-coding loci have been predicted (<http://www.phytozome.net/soybean#C>). With the completion of the soybean genome sequence, the full complement of TF-encoding genes from this important crop can be characterized and functionally analysed.

In this report, we searched for sequence-specific DNA-binding TFs using a prediction method which uses 51 Hidden Markov Models (HMMs) from the Pfam database. We also used 11 models, which were originally created by HMMbuild of HMMER2 package, to identify the domains within the putative TF proteins. The computational results predict that the soybean genome contains 5035 TF protein models coded from 4342 loci in 61 families. We created a database named 'SoybeanTFDB'. This database provides open access for researchers to all relevant and basic information on functional motifs, full-length cDNAs, promoter regions, genomic distribution, gene duplication and multiple sequence alignment of the DBDs for each TF family. Since most of these TFs have not been experimentally characterized for regulatory function as indicated by assessment in PubMed, we searched for their putative regulatory function by assessing annotations of the gene ontology (GO) using comparative analysis with their *Arabidopsis* counterparts. As a complement to this functional prediction using GO annotations, we also mapped all putative *cis*-regulatory elements that were documented within the PLACE database on all TF encoding genes. In this analysis, we placed a particular emphasis on abiotic stress responsive *cis*-elements. Knowledge gained from identifying the presence of stress responsive *cis*-elements, in addition to GO annotation, enables effective prediction of stress responsive TFs. Taken together, in this study, we demonstrate a comprehensive and high-quality census of TFs encoded within the soybean genome. These results provide a solid foundation for further systematic characterization of soybean TFs using traditional molecular approaches and/or genomic techniques at either the single-gene level or family-wide scale.

2. Materials and methods

2.1. Identification of TF repertoire in soybean

To identify TF encoding genes from the annotations of Glyma1 in the soybean genome, 51 HMMs of

Pfam³³ and those of 11 originally created using HMMbuild of the HMMER2 package (<http://hmmer.janelia.org/>) were applied, which corresponded to a total of 61 TF families (Supplementary Table S1). The modelled proteome data of annotated genes in Glyma1, which were downloaded from Phytozome (<http://www.phytozome.net/>), were subjected to a profile search for HMM dataset using Pfam-HMM with set thresholds of E -value, $E < 1e-5$ (Supplementary Table S1). The search results for each of the TF families were then applied to retrieve discovered regions as conserved DBDs and related annotations. To further classify genes with a conserved MYB domain into three subgroups: (R1)R2R3_MYB, MYB_related and atypical_MYB, the MYB soybean protein sequences were searched against previously classified *Arabidopsis* MYB genes³⁴ using blastp ($E < 1e-5$) and each top hit combination was applied to the classification. To avoid possible contaminations of pseudo response regulator or histidine kinase sequences into the GARP_ARRB family, genes containing CCT, CHASE, HATPase_c and HisKA together with Response_reg of Pfam domains were searched by InterProScan. Genes, which hit in this search, were subsequently removed from the GARP_ARRB family.

The putative TF encoding genes discovered in the soybean genome were classified into the following four categories based on their potential functionality as TFs. The first group of TFs (Category A) consists of TF encoding genes showing sequence identity $\geq 95\%$ and a blastn $E \leq 1e-100$ with GenBank soybean sequences having a functional description as TFs. Category A genes were classified with the highest confidence level after assessment with the PubMed database. The second group of TFs (Category B) is comprised TFs which have an equivalent protein domain arrangement (blastp $E \leq 1e-30$) for regulatory function in well-annotated plants, such as *Arabidopsis* and/or rice. The third group of TFs (Category C) combines possible TFs which show a significant hit with each of the HMM models used for DBD prediction (Pfam-HMM $E \leq 1e-20$). The last group contains TFs which have promiscuous HMM models with a threshold of settled E -values.

2.2. Structural and functional annotations for putative soybean TFs

For annotating TF encoding genes in soybean,³⁵ we used protein and cDNA sequences of soybean TFs as queries against the following protein and nucleotide datasets using the BLAST algorithm:³⁶ the nr protein DB of NCBI (<ftp://ftp.ncbi.nih.gov/blast/db>); the protein data presented in TAIR release 8 (ftp://ftp.arabidopsis.org/home/tair/Sequences/blast_datasets);

the protein data from UniProt (<http://www.uniprot.org/>); the TIGR/MSU Rice Genome Annotation Project (<http://rice.plantbiology.msu.edu/>) release 6; the soybean representative cDNA sequences in UniGene (<ftp://ftp.ncbi.nih.gov/repository/UniGene/>); the TIGR Transcript Assemblies (<http://plantta.jcvi.org/>); the Plant GDB (<http://www.plantgdb.org/>); the sequence sets of ESTs and high throughput cDNAs (HTCs) of RIKEN soybean full-length cDNA clones (<http://rsoy.psc.riken.jp/>);³⁷ the cDNAs of the previous version of the soybean genome annotation (Glyma0, Phytozome) and the target sequences of the Affymetrix soybean GeneChip (GPL4592 of NCBI GEO platform accession). All of the similarity searches using blastn were performed with threshold $E < 1e-100$, and the top scoring hit for each query was applied. All similarity searches with blastp against protein datasets were performed with a threshold $E < 1e-5$ to find possible functional descriptions for TF encoding genes. The top scoring hit for each query was applied.

Conserved domains in the protein sequence of putative TF encoding genes were identified with InterProScan and the InterPro DB (<http://www.ebi.ac.uk/interpro/>) to predict structures of DBD of TFs together with other functional domains and associated GO terms. All domains and those positions predicted by the search were retrieved and implemented them into our database. To determine the global characteristic features of functional categories of TF encoding genes of soybean, the TFs were assigned to possible GO terms based on a blastp similarity search to find *Arabidopsis* counterparts together with those annotated GOs of TAIR8. Particular emphasis was placed on sequences serving under the 'biological process' functional category.

TFs have been widely reported in plant TF databases such as DATF, AtTFDB, RARTF and PlnTFDB for *Arabidopsis* and DRTF, GRASIUS and PlnTFDB for rice. To annotate all putative soybean TFs in relation to *Arabidopsis* and rice counterparts, soybean TF sequences were assigned to annotation data related to TF families provided from each of the aforementioned databases based on sequence similarity searches between soybean proteome data and those of *Arabidopsis* and rice.^{8,14,38-42} The interrelated dataset of soybean genes, in combination with related *Arabidopsis* and rice TF annotations, were implemented into the SoybeanTFDB to provide cross references with other plant TFDBs.

2.3. Gene duplications and gene clusters in soybean TF families

Gene duplications and gene clusterings in soybean TF families were estimated by analysing the amino acid sequences of TF genes found on soybean

chromosomes. Specifically, the presence of gene pairs or gene clusters of closely homologous genes based on global sequence similarity with threshold of more than 60% amino acid sequence identity using cd-hit program of CD-HIT package were investigated.⁴³ Gene clusters are defined as genetic loci containing three or more closely homologous genes. Once identified, the pairs or gene clusters were used to assess the chromosomal allocation of highly homologous genes. Genes in tandem duplication are arbitrarily defined as those occurring within a sequence distance of 50 kb. On the other hand, genes that are duplicated in the same chromosome but reside >50 kb from each other are referred to as 'Duplications in same chromosome'. 'Duplications in different chromosomes' indicate pairs of highly homologous genes which reside on different chromosomes.

2.4. Discovery of *cis*-regulatory motifs in promoter regions of TF genes

To discover *cis*-regulatory motifs located in the promoter regions of each putative soybean TF gene and to investigate the enriched representation of *cis*-motifs in each TF family, *cis*-motif sequences from the PLACE database (version 30, 469 entries) (<http://www.dna.affrc.go.jp/PLACE/>)⁴⁴ and the stress responsive *cis*-motifs previously reported⁴⁵ were used as queries to search against the Glyma1 genome scaffold sequence using the fuzznuc program of EMBOSS package (<http://emboss.sourceforge.net/>). The results of pattern matches were subsequently assessed to identify matched sequences located on the -500, -1000 and -3000 bp upstream sequences from the putative transcription start site for each TF encoding gene defined in the Glyma1 annotation. The *cis*-element search results were implemented into the SoybeanTFDB as a searchable property. In addition, these search results were also incorporated as an annotation track of the genome browser (Gbrowse).

To assess the enrichment for the representative allocation of each *cis*-element identified on upstream sequences of each TF family, we analysed *cis*-element representations in the -1000 bp promoter region of TF members for TF families containing more than 50 gene loci to compare *cis*-element representations of randomly sampled gene loci of Glyma1. The computation of the overrepresentation test and its significance were performed by a Z-test as previously described.⁴⁶

2.5. Construction of a web accessible database

The database is implemented in MySQL and the web interface of Perl CGI and Java script run on the Apache Web server. The definition strings used for

sequence similarity searches for each database, the domain searches by InterProScan, *cis*-motif names from the PLACE database and the assigned GO terms have been assembled as a keyword database enabling users to specify queries on any keyword and to retrieve relevant information for genes from the SoybeanTFDB. A BLAST server was implemented to provide a similarity search interface for queried sequences using NCBI BLAST together with soybean Glyma1-related sequences, as well as those from *Arabidopsis* and rice. Generic Genome Browser (Gbrowse)⁴⁷ was also implemented in SoybeanTFDB with Glyma1 genome annotations released by Phytozome to visualize the gene annotations of the putative TF encoding genes together with *cis*-motifs found on the upstream sequence of the TF genes. All of the data in the SoybeanTFDB are accessible not only through a web interface but also as downloadable files from the website. The cross references of corresponding data for each of the entries were also implemented into the SoybeanTFDB together with the URLs for each of the original referenced data to provide hyperlinks on the web interface with seamless navigations.

3. Results and discussion

3.1. Identification of the soybean TF repertoire

For the purpose of identifying the repertoire of TFs within the soybean genome, we first define a class of proteins which bind DNA in a sequence-specific fashion. A protein is classified as a TF if it has a significant match to a model that we annotated as being a DBD, with the significance thresholds for HMM matches. Supplementary Table S1 summarized the HMMs used in TF predictions. For each HMM, we examined the description and associated literature to assess their sequence-specific DNA-binding capabilities. The pipeline that we used to predict soybean TFs began with retrieving the complete set of predicted proteins from the completely sequenced soybean genome. This approach was then followed by a HMMER search with all HMMs taken from the Pfam database (Fig. 1). In total, 4342 putative TF encoding loci which showed a significant match with these selected DBDs were extracted from the soybean genome sequence Glyma1 model (<http://www.phytozome.net/soybean#C>). These putative TFs represent 6.56% of the total number of predicted genes in soybean (Table 1 and Supplementary Table S2). In soy, this percentage of TFs to total gene number was similar to what has been observed for *Arabidopsis*. In the *Arabidopsis* genome, there are at least 1968 TFs which account for 7.23% of the total number of genes. Although the number of TFs

generally increase with the number of genes in a genome, interestingly the percentage of TF genes described in rice (3.68%) is less than expected (Table 1). The identified soybean TFs were classified into 61 families based on the presence of domains that were specific for the family (Table 2). Among the identified TFs, a significant proportion of the soybean TF repertoire has not been annotated with full-length open reading frames in the Glyma1

model. As a means to address this deficiency, we took advantage of our recently released soybean FL-cDNA collection of 4712 complete sequences and 68 661 ESTs to assess the Glyma1 annotation of the identified TFs (<http://www.legumebase.agr.miyazaki-u.ac.jp>).³⁷ Table 2 summarized the full-length information of the soybean TF encoding genes annotated by Glyma1 and the FL-cDNA collection. Detailed information for each gene is available on Supplementary Table S2. Next, we then grouped the TFs into four categories according to our confidence in their structure and functionality by assessing PubMed and relevant databases as described in Materials and methods (Fig. 2, and Supplementary Table S3). Relevant information of the soybean TF repertoire can be easily accessed at our website SoybeanTFDB (<http://soybeantfdb.psc.riken.jp>). Information that is readily available for the TF repertoire includes nucleotide and amino acid sequences, promoter regions and domain alignments within the family as well as multiple alignments with putative *Arabidopsis* and homologous rice genes.

Our prediction method depends heavily on the content of the Pfam database and the ability of the search algorithms to detect the DBDs in protein sequences, thus there are a few possible sources of inaccuracies in this prediction method. In addition, although the Glyma1 model contains more than 98% of known soybean protein-coding genes in its assembly, part of the TF repertoire may be clarified in the future by fine-tuning of the annotation. Finally, our literature analysis depends on the existing

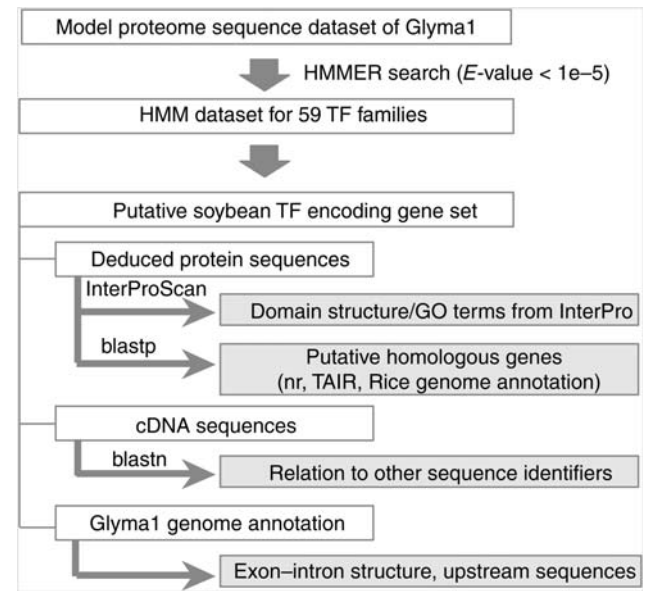


Figure 1. Schematic workflow of the computational pipeline used to discover and annotate genes encoding putative transcription factors in soybean.

Table 1. Numbers of TFs in *Arabidopsis*, rice and soybean

Species	No. of non-redundant TF gene loci ^a	No. of non-redundant gene loci ^b	Referenced annotation	Percentage of TFs ^c	Database	URL
<i>A. thaliana</i>	1922	27 235	TAIR8	7.06	DATF	http://datf.cbi.pku.edu.cn/
	1968			7.23	RARTF	http://rarge.gsc.riken.jp/rartf/
	1961			7.20	PlnTFDB	http://plntfdb.bio.uni-potsdam.de/v2.0/index.php?sp_id=ATH
	1737			6.38	AtTFDB	http://arabidopsis.med.ohio-state.edu/AtTFDB/
	1358			4.99	DBD	http://dbd.mrc-lmb.cam.ac.uk/DBD/index.cgi?About
<i>O. sativa</i>	1928	56 797	Rice Pseudomolecule Release 6	3.39	DRTF	http://drtf.cbi.pku.edu.cn/
	2095			3.69	PlnTFDB	http://plntfdb.bio.uni-potsdam.de/v2.0/index.php?sp_id=OSA
	2141			3.77	GRASSIUS	http://grassius.org/summary.html
	1629			2.87	DBD	http://dbd.mrc-lmb.cam.ac.uk/DBD/index.cgi?About
<i>G. max</i>	4342	66 210	Glyma1.0	6.56	SoybeanTFDB	http://soybeantfdb.psc.riken.jp

^aNumber of predicted non-redundant TF gene loci in each genome.

^bNumber of predicted non-redundant gene loci in each genome.

^cPercentage of TFs per genome.

Table 2. Characteristics of soybean transcription factors

TF gene families	No. of models ^a	No. of gene loci ^b	No. of models (FL) ^c	No. of models (not FL) ^d	No. of models assigned with RIKEN FL EST ^e	No. of models assigned with RIKEN FL-HTC ^f
1 (R1)R2R3_MYB	333	319	246	87	37	6
2 ABI3VP1	163	139	121	42	35	7
3 Alfin-like	27	18	26	1	7	2
4 AP2_EREBP	405	382	306	99	78	31
5 ARF	75	58	69	6	29	6
6 ARID	22	20	17	5	7	0
7 atypical_MYB	89	78	67	22	22	2
8 Aux_IAA	126	85	120	6	41	14
9 BBR-BPC	21	10	21	0	1	1
10 BES1	21	18	17	4	8	2
11 bHLH	390	325	317	73	72	18
12 bZIP	205	148	194	11	49	8
13 C2C2_Zn-CO-like	101	84	87	14	31	13
14 C2C2_Zn-Dof	87	81	78	9	13	5
15 C2C2_Zn-GATA	65	63	53	12	14	1
16 C2C2_Zn-YABBY	28	18	27	1	8	1
17 C2H2_Zn	270	258	211	59	42	5
18 C3H-Type1	178	151	154	24	50	14
19 CAMTA	14	14	12	2	4	0
20 CCAAT_Dr1	23	16	20	3	4	2
21 CCAAT_HAP2	42	23	40	2	9	2
22 CCAAT_HAP3	47	39	34	13	5	2
23 CCAAT_HAP5	26	23	20	6	6	0
24 CPP	22	17	17	5	1	0
25 E2F_DP	23	14	21	2	3	0
26 EIL	14	13	11	3	6	2
27 GARP_ARRB	21	21	16	5	4	2
28 GARP_G2-like	104	82	95	9	20	6
29 GeBP	17	17	6	11	4	1
30 GRAS	127	117	101	26	29	7
31 GRF	10	8	9	1	3	0
32 HB	283	242	240	43	67	20
33 HMG-box	50	26	43	7	18	3
34 HRT	1	1	1	0	1	1
35 HSF	65	59	54	11	11	6
36 JUMONJI	54	51	37	17	8	0
37 LFY	3	3	2	1	0	0
38 LIM	41	32	38	3	5	0
39 LUG	10	9	10	0	2	0
40 MADS	220	186	141	79	7	0
41 MBF1	3	3	3	0	2	1
42 MYB_related	168	135	138	30	35	8
43 NAC	205	187	159	46	37	11
44 Nin-like	23	23	17	6	2	0

Continued

Table 2. Continued

TF gene families	No. of models ^a	No. of gene loci ^b	No. of models (FL) ^c	No. of models (not FL) ^d	No. of models assigned with RIKEN FL EST ^e	No. of models assigned with RIKEN FL-HTC ^f
45 PcG	94	86	76	18	11	1
46 PHD	333	285	287	46	84	16
47 PLATZ	40	33	34	6	8	2
48 S1Fa-like	4	4	4	0	2	0
49 SAP	2	2	1	1	0	0
50 SBP	58	48	46	12	15	2
51 SRS	24	22	18	6	0	0
52 TCP	61	61	39	22	13	3
53 Trihelix	34	33	31	3	9	4
54 TUB	37	24	33	4	18	2
55 ULT	32	32	5	27	0	0
56 VOZ	8	7	7	1	2	1
57 Whirly	11	7	11	0	2	0
58 WRKY_Zn	219	198	167	52	47	12
59 zf-HD	57	56	37	20	4	1
60 zf-TAZ	8	8	6	2	1	0
61 ZIM	57	34	55	2	28	13
Total	5035	4342	4032	1003	1003	249

^aNumber of predicted TF models in Glyma1 model.
^bNumber of predicted TF loci in Glyma1 model.
^cNumber of predicted full-length TF models in Glyma1 model.
^dNumber of predicted not full-length TF models in Glyma1 model.
^eNumber of predicted full-length TF models in soybean assigned with RIKEN full-length ESTs.
^fNumber of predicted full-length TF models in soybean assigned with RIKEN full-length high throughput cDNAs.



Figure 2. The distributions of soybean TF encoding genes are classified into four categories of annotation levels. Category A includes soybean gene models showing sequence identity $\geq 95\%$ and a blastn $E \leq 1e-100$ with GenBank soybean sequences having a functional description as TFs. Category B includes gene models which have an equivalent protein domain arrangement (blastp $E \leq 1e-30$) for regulatory function in well-annotated plants, such as *Arabidopsis* and/or rice. Category C includes gene models which show a significant hit with each of the HMMs used for DBD prediction (Pfam-HMM $E \leq 1e-20$). Category D includes TF genes which have promiscuous HMMs with a threshold of settled E -values.

available published information pertaining to each gene, which will need to be updated as new findings are reported. The availability of updated HMM libraries or refinements of existing ones and better fine-tuned annotation and continuous searches for newly reported literature will enable us to improve

the TF prediction coverage. We will continue to update the website with new information when it becomes available.

Literature analysis, which is achieved by assessing corresponding genes deposited as soybean TF genes in the GenBank core nucleotide division together with associated identifiers of PubMed, has revealed that the majority of soybean TFs remains experimentally uncharacterized. Thus, we attempted to further extend our current knowledge base regarding their regulatory function by assessing the putative functions of soybean TFs via comparative analyses with relevant GO annotations of *Arabidopsis* in TAIR. First, we analysed the profile of GO terms at the biological process level which could be assigned to soybean TFs based on sequence similarity searches against *Arabidopsis* counterparts having GO terms in TAIR. In order to grasp the overall representation of GO terms in applied entries of soybean TFs, all of the assigned terms were counted after the similarity searches were completed. With the exception of ‘regulation of transcription’, ‘DNA binding’ and ‘biological process’, the top 21 most abundant terms were

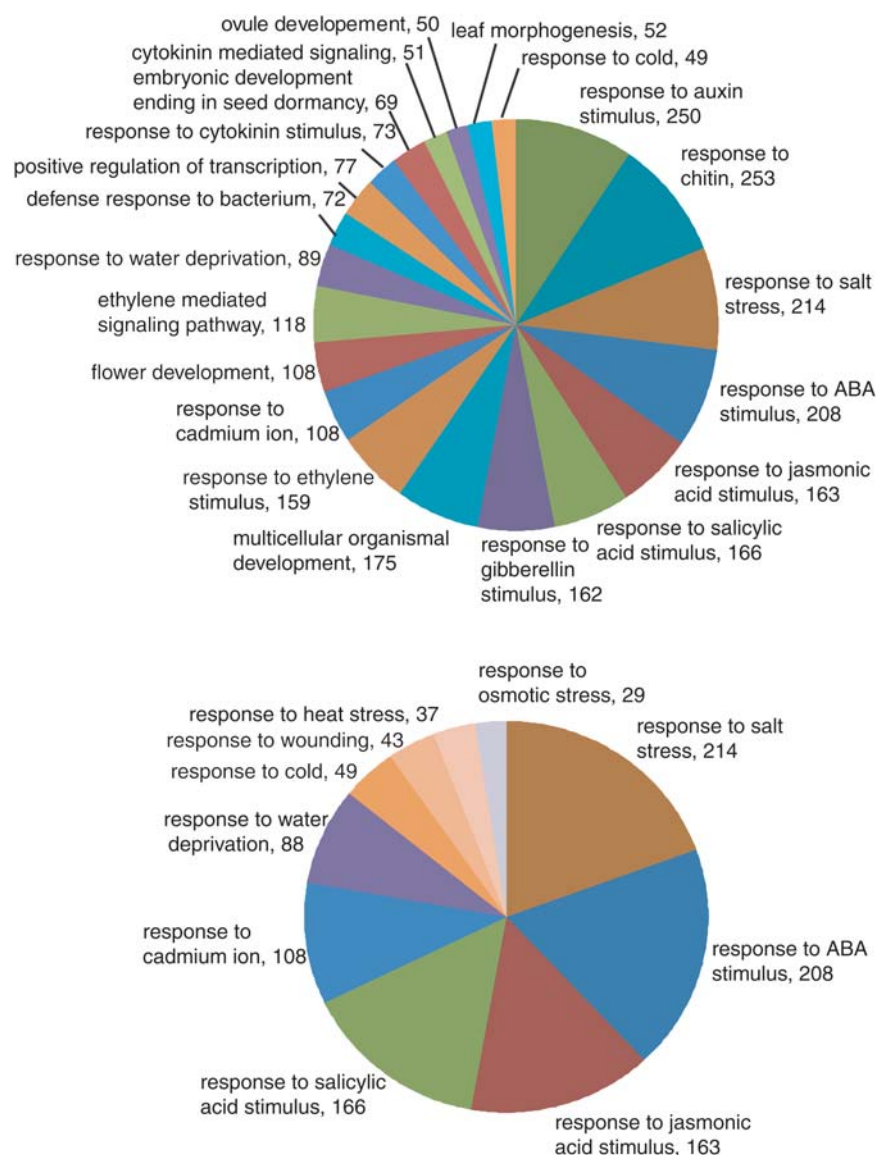


Figure 3. The representative distributions of the GO terms for biological processes associated with soybean TF encoding genes. The top 21 abundantly found GO terms were assigned based on homology searches against annotated *Arabidopsis* genes (A). Abundant distribution of TFs in GO terms related to the response to various types of abiotic stresses in the soybean TF dataset (B). Gene numbers are displayed next to the terms.

subsequently used to classify the TFs. The contig results of these total 21 GO terms for each soybean TF, which was based on similarity with *Arabidopsis* TFs, are provided in Supplementary Table S4. Figure 3A illustrates the distribution of soybean TFs in the 21 most abundant GO terms. A significant proportion of soybean TFs are related to stress and hormone responses (Fig. 3B), indicating the important role of TFs in controlling these biological processes. Of these assigned regulatory functions, responses to auxin, chitin and salt stress are the most highly represented. It is acknowledged that these annotations are the first steps in functional prediction, and researchers must use original publications as a source for a higher level of detailed

information. In addition, it is ideal if functional analyses can be performed in order to gain a detailed understanding of gene function. Overall, these analyses emphasize the limited amount of functional information that we know regarding the biological processes that most of the TFs mediate, even for model plants such as *Arabidopsis*. Directing research efforts into uncharacterized TFs—for example, using high-throughput genomic surveys to describe the key features combined with a detailed examination using traditional molecular approaches—will undoubtedly accelerate our functional understanding of these important regulatory genes. The NAC TF family, which is widely distributed in plants but so far has not been found in other eukaryotes, is an

excellent example of how research interests can suddenly arise the following key findings. The acceleration in functional studies has revealed their diverse functions in different biological aspects and future follow-up studies will rapidly improve our understanding of the regulatory function of NAC members. A greater understanding of how TFs operate will be subsequently translated into their potential applications to enhance plant productivity.^{4,48,49}

3.2. Structural feature of TFs

As mentioned above, the most common classification of TFs is based on the structure of their DBD.^{7,14} Grouping TFs by their structural domains has been extremely useful in gaining insights into how they recognize and bind specific DNA sequence. This strategy has also been proven successfully for characterizing their evolutionary histories as well. Moreover, the DBD may provide clues to their biological function. For example, ABI3/VP1 TFs are often associated with the regulation of abscisic acid (ABA)-responsive genes during seed development.⁵⁰

Since structural features of TF families have been extensively characterized in other reports, we do not cover this in detail within this report.⁸ However, it is worthy to note that soybean contains a number of large families which consist of more than 100 members (Supplementary Table S5). For example, the large AP2_EREBP family alone contains 405 TF models and accounts for a total of 8.04% of the TF repertoire. The bHLH and (R1)R2R3_MYB TFs also represent major families with 390 and 333 members, respectively, which together occupy 14.36% of the TF repertoire. These observations agree with the previous studies in *Arabidopsis* and rice, which confirmed that the same three families contain the highest numbers of TFs in these model systems (Supplementary Table S5). In addition, the plant-specific NAC family, which comprises 205 models in soybean, represents a similar ratio in *Arabidopsis* and rice (Supplementary Table S5). Taken together, these results suggest a similar tendency in the evolution of major TF families in plants. Furthermore, given that the size of TF families is influenced in part by the number of different DNA sequences that they are able to recognize, the DBDs of AP2_EREBP, bHLH and (R1)R2R3_MYB TF families may be able to diversify their collection of target sequences. As a result, they occur in the greater numbers in a genome.^{4,51,52}

3.3. Chromosomal distribution and gene duplications of TFs

Our analysis has indicated that the soybean TF families are scattered throughout the genome. The

larger families, such as AP2_EREBP, (R1)R2R3_MYB, have members that are distributed on every chromosome in soybean (Supplementary Table S2). The local distribution of TF genes relative to each other is also of interest. Previous studies have described duplications and clusters of highly homologous genes. In *Arabidopsis*, tandem gene duplications and large-scale duplications on different chromosomes may account for >60% of the genome.⁷ In soybean, we were able to distinguish between two types of duplications and clusters based on the evolutionary history of the TF-coding genes that they contain. The first type of duplications and clusters consists of a series of paralogous genes, suggesting that they arose through repeated tandem duplications which originated from a founding locus. In contrast, the second type of duplications and clusters is not comprised paralogous genes. We anticipate that the TF-coding genes in these duplications and clusters arose independently of each other at diverse locations within the genome. Over time, it is likely that they relocated to form these duplications and clusters. Table 3 summarizes gene duplications and gene clusters in soybean TF families. Closely related genes, which are defined by >60% amino acid sequence identity, account for ~77.75% of the total number in the TF families (Table 3). Pairs of duplicated genes on different chromosomes are most common and gene clusters of three or more highly related genes are also widely found (Table 3). On the basis of the distance of their occurrence, a few of the duplicated genes could be classified arbitrarily as either genes that were duplicated on same chromosome or genes that were tandemly duplicated. Evolutionary studies and haploid genome analysis have suggested that the soybean genome experienced a tetraploidization event which occurred an estimated 10–15 million years ago. Since then, the soybean genome has gone through extensive gene rearrangements and deletions to become diploidized.⁵³ Therefore, we can observe in soybean that multigene families, including TF families, contain highly related genes.^{24,54}

3.4. Promoter regions of the TFs and the discovery of cis-elements in the TF promoter regions

Cis-regulatory elements, which are the binding sites for TFs located in the promoter regions of genes, are the functional elements that determine the timing and location of transcriptional activity. Over the years, extensive promoter analyses have identified a large number of cis-elements, which are important molecular switches involved in the transcriptional regulation of a dynamic network of gene activities controlling various biological processes such as

Table 3. Classification of homologous soybean TF genes

TF family	No. of gene loci ^a	No. of genes with close homolog ^b	Percentage of genes with close homolog	No. of individual genes	Duplications in different chromosomes ^c	Duplications in same chromosome ^d	Tandem duplications ^e	No. of gene clusters/no. of genes in cluster (no. of chromosomes) ^f
(R1)R2R3_MYB	318	261	82.08	57	50	2	1	43/155 (20)
ABI3VP1	139	90	64.75	49	20	1	2	12/44 (17)
Alfin-like	18	18	100.00	0	1	0	0	1/16 (11)
AP2_EREBP	381	309	81.10	72	74	1	0	42/159 (20)
ARF	58	45	77.59	13	8	1	0	7/27 (16)
ARID	20	9	45.00	11	1	0	0	2/7 (6)
atypical_MYB	78	42	53.85	36	14	1	1	3/10 (6)
Aux_IAA	85	72	84.71	13	9	0	0	13/54 (15)
BBR-BPC	10	10	100.00	0	2	0	0	1/6 (4)
BES1	18	17	94.44	1	3	0	0	2/11 (9)
bHLH	323	269	83.28	54	63	1	2	38/137 (20)
bZIP	148	124	83.78	24	26	1	0	18/70 (20)
C2C2_Zn-CO-like	84	67	79.76	17	19	0	0	7/29 (13)
C2C2_Zn-Dof	81	60	74.07	21	17	1	0	7/24 (13)
C2C2_Zn-GATA	63	53	84.13	10	11	0	0	9/31 (15)
C2C2_Zn-YABBY	18	15	83.33	3	2	0	0	3/11 (8)
C2H2_Zn	257	187	72.76	70	61	2	2	15/57 (18)
C3H-TypeI	151	123	81.46	28	25	0	1	15/71 (17)
CAMTA	14	12	85.71	2	6	0	0	0/0 (0)
CCAAT_Dr1	16	14	87.50	2	2	0	0	3/10 (9)
CCAAT_HAP2	23	16	69.57	7	2	0	0	3/12 (10)
CCAAT_HAP3	39	32	82.05	7	4	0	0	4/24 (14)
CCAAT_HAP5	23	19	82.61	4	3	0	1	3/11 (9)
CPP	17	10	58.82	7	3	0	0	1/4 (4)
E2F_DP	14	11	78.57	3	2	0	0	2/7 (5)
EIL	13	12	92.31	1	2	0	0	2/8 (6)
GARP_ARRB	20	13	65.00	7	3	0	0	2/7 (6)
GARP_G2-like	82	59	71.95	23	18	0	0	7/23 (13)
GeBP	17	13	76.47	4	2	0	0	2/9 (8)
GRAS	117	102	87.18	15	23	0	0	14/56 (20)
GRF	8	2	25.00	6	1	0	0	0/0 (0)
HB	240	197	82.08	43	31	0	0	33/135 (20)
HMG-box	26	22	84.62	4	5	0	0	2/12 (10)
HRT	1	0	0.00	1	0	0	0	0/0 (0)
HSF	59	48	81.36	11	12	0	0	7/24 (17)
JUMONJI	51	28	54.90	23	8	0	1	3/10 (9)
LFY	3	2	66.67	1	1	0	0	0/0 (0)
LIM	32	28	87.50	4	1	0	0	5/26 (15)
LUG	9	8	88.89	1	2	0	0	1/4 (4)
MADS	186	154	82.80	32	25	2	1	18/98 (19)
MBF1	3	3	100.00	0	0	0	0	1/3 (2)
MYB_related	135	112	82.96	23	17	1	1	16/74 (19)

Continued

Table 3. Continued

TF family	No. of gene loci ^a	No. of genes with close homolog ^b	Percentage of genes with close homolog	No. of individual genes	Duplications in different chromosomes ^c	Duplications in same chromosome ^d	Tandem duplications ^e	No. of gene clusters/no. of genes in cluster (no. of chromosomes) ^f
NAC	187	173	92.51	14	26	0	1	30/119 (20)
Nin-like	23	15	65.22	8	4	0	0	2/7 (4)
PcG	86	56	65.12	30	21	0	0	4/14 (9)
PHD	285	188	65.96	97	51	1	0	18/84 (20)
PLATZ	33	27	81.82	6	2	0	1	4/21 (13)
S1Fa-like	4	4	100.00	0	0	0	0	1/4 (4)
SAP	2	2	100.00	0	1	0	0	0/0 (0)
SBP	48	39	81.25	9	9	0	0	6/21 (13)
SRS	22	18	81.82	4	4	0	0	3/10 (8)
TCP	61	44	72.13	17	19	0	0	2/6 (3)
Trihelix	33	25	75.76	8	11	0	0	1/3 (3)
TUB	24	20	83.33	4	2	0	0	3/16 (12)
ULT	24	20	83.33	4	1	1	0	2/16 (1)
VOZ	7	7	100.00	0	1	0	0	1/5 (4)
Whirly	7	6	85.71	1	3	0	0	0/0 (0)
WRKY_Zn	198	166	83.84	32	49	0	2	17/64 (20)
zf-HD	56	46	82.14	10	7	0	1	6/30 (15)
zf-TAZ	8	5	62.50	3	1	0	0	1/3 (3)
ZIM	34	27	79.41	7	8	0	0	3/11 (9)

^aNumber of predicted TF loci found in soybean chromosomes (Glyma1 model).

^bGenes were considered closely homologs if they showed >60% amino acid sequence identity.

^cPairs of closely homologous genes which are duplicated in different chromosomes.

^dPairs of closely homologous genes which are duplicated in same chromosome but resided >50 kb apart from each other.

^ePairs of closely homologous genes which are duplicated in same chromosome but resided <50 kb apart from each other.

^fClusters of three or more closely homologous genes.

abiotic stress responses, hormone responses and developmental processes.^{45,55} The PLACE database (<http://www.dna.affrc.go.jp/PLACE/>) has consolidated all of the published *cis*-motifs that have been identified to date. In addition, a number of stress responsive *cis*-motifs are also reported, which are of great interest to our area of research.⁴⁵ To facilitate the functional characterization of soybean TFs, we retrieved the promoter regions for all of the TF genes from soybean genomic sequence database. Specifically, we retrieved 500, 1000 and 3000 bp of sequence upstream from putative transcription start sites. We provided this data on our website in addition to other relevant information on the TFs for convenient downloading. The −500, −1000 and −3000 bp promoter regions were subjected to an extensive *in silico* analyses to search for the existence of all putative known *cis*-regulatory motifs. In addition, we also analysed the enrichment of all of the *cis*-motifs in each TF family using −1000 bp promoter regions as described in Materials and methods.

Information on the *cis*-elements located in the promoter region of each TF is accessible on the detailed page of each TF gene under '*cis*-motif prediction' function (Fig. 4C). In addition, our website provides the '*cis*-motif (PLACE)' search function, which enables the search for all types of *cis*-motifs provided by the PLACE database in promoter region of any TF and/or the search for those TFs which contains the *cis*-motif(s) of interest (Fig. 4A). In combination with GO annotations (Fig. 4H), these data can facilitate the systematic functional predictions of soybean TFs.

Numerous *cis*-elements have been reported for their essential roles in determining the tissue-specific or stress-induced expression patterns of genes.^{45,55} Recently, a systematic combinatorial *in silico* analysis of *cis*-motifs and expression patterns in *Arabidopsis* indicated a positive correlation between multi-stimuli response genes and *cis*-element density in upstream regions.⁵⁶ Inspection of the relationship of the existence of *cis*-regulatory elements and the expression patterns of the TF genes can therefore



Figure 4. The web-based user interface of SoybeanTFDB and a demonstration of a typical example of related annotations for a putative soybean TF encoding gene. The interface of SoybeanTFDB provides search queries for the names of TF families, keywords, sequence identifiers, identifiers of domains supported by InterProScan, GO terms and available *cis*-motifs (A). The search results are listed for each of the TF families with a description of corresponding genes based on similarity searches (B). Users are able to navigate to the detailed annotation pages to browse the related annotations. The detailed annotation pages provide summarized basic information on each of the gene models annotated in Glyma1 with gene structure. The figure for a gene structure is accessible via a hyperlink to a genome browser which is browsed together with other sequences allocated onto the soybean genome (C). The sequences of cDNAs and proteins are provided and all clickable buttons navigate users to the blast search interface directory (D). The similarity search results for each of the entries against NCBI nr, gene models of *Arabidopsis* and rice with detailed search results and hyperlinks to the original data (E). Resultant hierarchical clustering of homologous soybean TF genes can be browsed with multiple alignment of each cluster (F). The identifiers assigned provide hyperlinks to the annotation web pages of *Arabidopsis* and rice TF databases. Information of other sequence identifiers for representative transcript sequence databases, including PlantGDB, UniGene and TIGR GI, as well as the probe ID of target sequences on the soybean Affymetrix GeneChip, are also accessible. Furthermore, the interface also provides corresponding hyperlinks to the FL-cDNA provided from RIKEN (<http://rsoy.psc.riken.jp/>) (G). The GO terms assigned to each of the entries based on InterProScan and sequence similarity search against the annotated genes of *Arabidopsis* and TAIR8 (H). The domain structure predicted by InterProScan is provided (I). The result of a *cis*-motif sequence pattern search of promoter regions for each gene is shown together with genomic gene structure (J).

help predict the function of the respective TFs during development, in different organs, cell types or in response to various endo- or exogenous stimuli. Additionally, quantitative models that describe how combinations of *cis*-elements dictate changes in expression will play an important role for enriching our understanding of the transcriptional response of individual genes to environmental perturbations.²⁶

3.5. *Cis-element- and comparative sequence analysis-based prediction of abiotic stress responsive TFs in soybean*

Plants respond to environmental changes by altering large-scale transcriptional responses. The exquisite sensitivity and specificity of these responses are controlled in large part by the *cis*-regulatory elements. The molecular mechanisms regulating gene expression in response to abiotic stresses have been studied by analysing the *cis*- and *trans*-acting elements, i.e. the sequence-specific binding TFs.^{4,45}

Genes induced by stresses are classified into two groups: functional genes and regulatory genes. The regulatory group includes genes encoding various TFs which can regulate various stress-inducible genes cooperatively or separately and may constitute gene networks. Identification and functional analysis of these stress-inducible TFs should provide more information on the complex regulatory gene networks that are involved in stress responses. At the present time, the functions for most of stress responsive TF encoding genes are not fully understood. Some of the stress-inducible TFs have been overexpressed in transgenic plants and result in stress-tolerant phenotypes.^{4,45,49}

Recent studies have substantiated that sequence similarity-based clustering of the members of several TF families correlates with their function. Phylogenetic analysis of the AP2_EREBP and NAC families of soybean and the rice NAC family with orthologs from other plant species whose stress responsive expression pattern and/or function are known, resulted in a nearly perfect match between sequence conservation and function or expression patterns. These similarities clearly demonstrate that this can serve as a reliable approach to rationalize systematic functional predictions of different TF families.^{21,24,54} Moreover, increasing evidence indicates that the *cis*-motifs are highly conserved among orthologous or paralogous genes and coregulated genes and defined *cis*-elements can effectively aid in the genome-wide screening of ABA and abiotic stress responsive genes.^{57–59} These observations together prompted us to investigate in a comprehensive fashion the relationship between TFs and abiotic stress with the integration of *cis*-element annotation

and comparative sequence analysis using stress responsive GO terms which aimed to identifying soybean TFs which may function in abiotic stress response. We, therefore, carried out comparative sequence analysis with stress-responsive *Arabidopsis* TFs to predict the soybean TFs with stress responsive GO terms (Fig. 3B). We also characterized information on stress-responsive *cis*-element distributions in promoter regions of each soybean TF gene on our webpage for querying and searching for putative stress responsive TFs in each family using 'cis-motif (stress responsive)' search function. With the help of our soybean TF database, we can use, for example, the 'cis-motif (stress responsive)' search function to identify TF genes which harbour major known stress responsive *cis*-motif(s) in their promoter regions (Fig. 4A). Next, we screen the identified TFs using GO annotation provided for each TF on detailed annotation page (Fig. 4H). Thus, we will be able to identify the putative stress responsive TFs based on both the existence of stress responsive *cis*-motif(s) and the associated stress responsive GO terms. The predicted stress responsive function of the identified TFs shall be then confirmed by experiments. The existence of major stress responsive *cis*-motifs enriched in –1000 bp promoter regions for a number of TF families was summarized in Table 4.

3.6. *RIKEN soybean TF database*

We constructed a TF database named SoybeanTFDB which is based on the identified soybean TF repertoire. Access to our database is available via the following link: <http://soybeantfdb.psc.riken.jp>, and all of the data described above are available for viewing and immediate downloading. The scientific community can browse predictions for a total of 5035 TF models and receive classifications for submitted nucleotide and protein sequences. Multiple alignments of amino acid sequences within TF families are also available for downloading and can be used for the construction of phylogenetic trees. We also provided clustered results showing amino acid similarity with different levels of amino acid identity (30, 60 and 90%), search functions for functional motif information of InterProScan, *cis*-motifs in promoter regions of TFs and GO annotations. Furthermore, cross-references and links to other databases such as *Arabidopsis* TAIR8, TIGR rice, UniProt, SoyBase, soybean FL-cDNA and other TF databases such as AtTFDB, DATE, RARTE, DRTE, Grassius, PlnTFDB are available. On the first page of SoybeanTFDB, we provide four types of search keywords to find an entry: 'TF search', 'Similarity search', 'Genome browser' and 'Quick search'. Similarity search allows search using either nucleotide or

Table 4. Enriched *cis*-regulatory motifs found in promoter region (1000 bp upstream from transcription start site of each gene) of genes encoding each of the TF families

<i>Cis</i> -motif name ^a	<i>Cis</i> -motif pattern ^a	TF family	No. of gene loci hit	No. of gene loci	Mean observed ^b	Mean expected ^c	Z-score	<i>P</i> -value (<0.001)
ABRE1	[TC]ACGTGGC	C2C2_Zn-CO-like	4	84	47.385	14.074	8.81	0
		C2C2_Zn-GATA	3	63	47.784	14.074	8.92	0
		C3H-Type1	6	151	39.851	14.074	6.82	4.577E-12
		JUMONJI	2	51	39.435	14.074	6.71	9.7873E-12
		NAC	5	187	26.593	14.074	3.31	0.00046339
		TCP	5	61	82.433	14.074	18.08	0
ABRE2	ACGTG[GT]C	WRKY_Zn	6	198	29.952	14.074	4.20	0.000013318
		(R1)R2R3_Myb	26	319	81.996	55.79	3.59	0.00016627
		AP2_EREBP	31	382	81.157	55.79	3.47	0.00025671
		Aux_IAA	7	85	82.312	55.79	3.63	0.00014072
		C2C2_Zn-CO-like	13	84	154.522	55.79	13.52	0
		C2C2_Zn-Dof	10	81	123.231	55.79	9.24	0
		C2C2_Zn-GATA	5	63	79.277	55.79	3.22	0.00064947
		C3H-Type1	12	151	79.117	55.79	3.19	0.00070084
		JUMONJI	7	51	137.741	55.79	11.22	0
		MADS	18	186	96.524	55.79	5.58	1.2169E-08
		NAC	27	187	144.379	55.79	12.13	0
		TCP	8	61	131.2	55.79	10.33	0
		Atypical_MYB	10	78	128.071	55.79	9.90	0
CE1	TGCCACCGG	C2H2_Zn	1	258	3.952	0.571	4.39	5.7069E-06
		JUMONJI	1	51	19.698	0.571	24.83	0
		WRKY_Zn	4	198	20.174	0.571	25.44	0
CRT	GGCCGACAT	AP2_EREBP	1	382	2.538	0.324	3.88	0.000051901
		C2H2_Zn	1	258	3.86	0.324	6.20	2.8371E-10
DRE	TACCGACAT	ABI3VP1	1	139	7.255	0.77	7.36	9.0372E-14
		AP2_EREBP	3	382	7.931	0.77	8.13	2.2204E-16
		NAC	1	187	5.351	0.77	5.20	9.9255E-08
ICer2	ACTCCG	AP2_EREBP	22	382	57.453	37.698	3.25	0.00057027
		C2C2_Zn-GATA	4	63	63.223	37.698	4.20	0.000013136
		JUMONJI	4	51	78.537	37.698	6.73	8.7457E-12
		PHD	20	285	70.006	37.698	5.32	5.1702E-08
		TCP	4	61	65.403	37.698	4.56	2.5263E-06
MYBR	TGTTAG	C2H2_Zn	19	258	73.778	48.032	3.90	0.000047443
		MADS	13	186	69.508	48.032	3.26	0.00056509
		TCP	5	61	82.123	48.032	5.17	0.000000118
MYCR	CACATG	(R1)R2R3_Myb	98	319	307.51	230.595	5.93	1.5169E-09
		ABI3VP1	41	139	295.074	230.595	4.97	3.3303E-07
		AP2_EREBP	112	382	293.307	230.595	4.83	6.6647E-07
		ARF	19	58	327.992	230.595	7.51	2.9865E-14
		Aux_IAA	23	85	270.693	230.595	3.09	0.00099623
		C2C2_Zn-CO-like	25	84	298.315	230.595	5.22	8.9042E-08
		C2C2_Zn-Dof	29	81	358.556	230.595	9.87	0
		C2H2_Zn	70	258	271.297	230.595	3.14	0.00085076
		Myb_related	38	135	281.165	230.595	3.90	0.000048357
		bZIP	42	148	283.077	230.595	4.05	0.000026039
NACR	ACACGCATGT	C2H2_Zn	1	258	3.823	0.494	4.93	4.1633E-07
		WRKY_Zn	1	198	4.944	0.494	6.59	2.2463E-11
		bZIP	1	148	6.804	0.494	9.34	0

^aAccording to Yamaguchi-Shinozaki *et al.*⁴⁵^bThe mean values observed were calculated by counting motif pattern hit in 1000 random samplings in each 1000 trials for promoter pools of each TFs.^cThe mean values expected were calculated by counting motif patterns hit in 1000 random samplings in each 1000 trials for promoter pools of all genes annotated in soybean genome.

amino acid sequence of any TF gene. Genome browser enables search using gene IDs and Quick search allows search function via any essential keywords such as 'NAC'. Within the TF search keyword,

we provide seven search functions (Fig. 4A). Figure 4 illustrates the web-based user interface of SoybeanTFDB with a detailed description. One can easily carry out functionality predictions for any TF

of interest based on GO annotations and *cis*-motif search results. For instance, putative abiotic stress responsive TFs can be searched based on the existence of stress responsive *cis*-motifs and GO annotations. Thus, the database that we have developed consolidates comprehensive information for all of the members of soybean TF repertoire. This database is a very user-friendly interface which aims to meet the broad demands of researchers who strive to perform research with soybean TFs with the goal of gaining greater understanding of their putative roles in plant development, differentiation and environmental responses. Taken together, SoybeanTFDB will serve as an *in silico* analysis-based basic platform for the elucidation of regulatory mechanisms underlying different developmental and physiological processes and stress responses. We strongly feel that this database will rapidly accelerate the progress in 'transcription factoromics' of soybean, comparative genomics of TF repertoires both within legume species and between legumes and other species, as well as facilitate genetic engineering programs to improve the productivity of soybean grown in adverse conditions.

Acknowledgements: The soybean sequence data were produced by the US Department of Energy Joint Genome Institute <http://www.jgi.doe.gov/> in collaboration with the scientific user community.

Supplementary data: Supplementary Data are available at www.dnaresearch.oxfordjournals.org.

Funding

Funding support from Grants-in-Aid (Start-up) for Scientific Research, Ministry of Education, Culture, Sports, Science, and Technology of Japan (No. 21870046) is gratefully appreciated.

References

1. Riechmann, J.L. and Ratcliffe, O.J. 2000, A genomic perspective on plant transcription factors, *Curr. Opin. Plant Biol.*, **3**, 423–34.
2. Czechowski, T., Bari, R.P., Stitt, M., Scheible, W.R. and Udvardi, M.K. 2004, Real-time RT-PCR profiling of over 1400 *Arabidopsis* transcription factors: unprecedented sensitivity reveals novel root- and shoot-specific genes, *Plant J.*, **38**, 366–79.
3. Tran, L.-S.P., Nakashima, K., Shinozaki, K. and Yamaguchi-Shinozaki, K. 2007, Plant gene networks in osmotic stress response: from genes to regulatory networks, *Methods Enzymol.*, **428**, 109–28.
4. Yamaguchi-Shinozaki, K. and Shinozaki, K. 2006, Transcriptional regulatory networks in cellular responses and tolerance to dehydration and cold stresses, *Annu. Rev. Plant Biol.*, **57**, 781–803.
5. Bustamante, C.D., Fledel-Alon, A., Williamson, S., et al. 2005, Natural selection on protein-coding genes in the human genome, *Nature*, **437**, 1153–7.
6. Carroll, S.B. 2008, Evo-devo and an expanding evolutionary synthesis: a genetic theory of morphological evolution, *Cell*, **134**, 25–36.
7. Riechmann, J.L., Heard, J., Martin, G., et al. 2000, *Arabidopsis* transcription factors: genome-wide comparative analysis among eukaryotes, *Science*, **290**, 2105–10.
8. Riaño-Pachón, D.M., Ruzicic, S., Dreyer, I. and Mueller-Roeber, B. 2007, PlnTFDB: an integrative plant transcription factor database, *BMC Bioinformatics*, **8**, 42.
9. Guo, A.Y., Chen, X., Gao, G., et al. 2008, PlantTFDB: a comprehensive plant transcription factor database, *Nucleic Acids Res.*, **36**, D966–9.
10. Wilson, D., Charoensawan, V., Kummerfeld, S.K. and Teichmann, S.A. 2008, DBD-taxonomically broad transcription factor predictions: new content and functionality, *Nucleic Acids Res.*, **36**, D88–92.
11. Cherry, J.M., Adler, C., Ball, C., et al. 1998, SGD: *Saccharomyces* genome database, *Nucleic Acids Res.*, **26**, 73–9.
12. Perez-Rueda, E. and Collado-Vides, J. 2000, The repertoire of DNA-binding transcriptional regulators in *Escherichia coli* K-12, *Nucleic Acids Res.*, **28**, 1838–47.
13. Gray, P.A., Fu, H., Luo, P., et al. 2004, Mouse brain organization revealed through direct genome-scale TF expression analysis, *Science*, **306**, 2255–7.
14. Iida, K., Seki, M., Sakurai, T., et al. 2005, RARTF: database and tools for complete sets of *Arabidopsis* transcription factors, *DNA Res.*, **12**, 247–56.
15. Reece-Hoyes, J.S., Deplancke, B., Shingles, J., Grove, C.A., Hope, I.A. and Walhout, A.J. 2005, A compendium of *Caenorhabditis elegans* regulatory transcription factors: a resource for mapping transcription regulatory networks, *Genome Biol.*, **6**, R110.
16. Adryan, B. and Teichmann, S.A. 2006, FlyTF: a systematic review of site-specific transcription factors in the fruit fly *Drosophila melanogaster*, *Bioinformatics*, **22**, 1532–3.
17. Moreno-Campuzano, S., Janga, S.C. and Perez-Rueda, E. 2006, Identification and analysis of DNA-binding transcription factors in *Bacillus subtilis* and other *Firmicutes*—a genomic approach, *BMC Genomics*, **7**, 147.
18. Park, J., Park, J., Jang, S., et al. 2008, FTFD: an informatics pipeline supporting phylogenomic analysis of fungal transcription factors, *Bioinformatics*, **24**, 1024–5.
19. Udvardi, M.K., Kakar, K., Wandrey, M., et al. 2007, Legume transcription factors: global regulators of plant development and response to the environment, *Plant Physiol.*, **144**, 538–49.
20. Tran, L.-S.P., Nakashima, K., Sakuma, Y., et al. 2004, Isolation and functional analysis of *Arabidopsis* stress-inducible NAC transcription factors that bind to a drought-responsive *cis*-element in the *early responsive to dehydration stress 1* promoter, *Plant Cell*, **16**, 2481–98.
21. Fang, Y., You, J., Xie, K., Xie, W. and Xiong, L. 2008, Systematic sequence analysis and identification of

- tissue-specific or stress-responsive genes of NAC transcription factor family in rice, *Mol. Genet. Genomics*, **280**, 535–46.
22. Liao, Y., Zou, H.F., Wei, W., et al. 2008, Soybean *GmbZIP144*, *GmbZIP162* and *GmbZIP178* genes function as negative regulator of ABA signaling and confer salt and freezing tolerance in transgenic *Arabidopsis*, *Planta*, **228**, 225–40.
 23. Zhou, Q.Y., Tian, A.G., Zou, H.F., et al. 2008, Soybean WRKY-type transcription factor genes, *GmWRKY13*, *GmWRKY21*, and *GmWRKY54*, confer differential tolerance to abiotic stresses in transgenic *Arabidopsis* plants, *Plant Biotechnol. J.*, **6**, 486–503.
 24. Tran, L.-S.P., Quach, T.N., Guttikonda, S.K., et al. 2009, Molecular characterization of stress-inducible *GmNAC* genes in soybean, *Mol. Genet. Genomics*, **281**, 647–64.
 25. Gong, W., Shen, Y.P., Ma, L.G., et al. 2004, Genome-wide ORFeome cloning and analysis of *Arabidopsis* transcription factor genes, *Plant Physiol.*, **135**, 773–82.
 26. Gertz, J. and Cohen, B.A. 2009, Environment-specific combinatorial cis-regulation in synthetic promoters, *Mol. Syst. Biol.*, **5**, 244.
 27. Vaquerizas, J.M., Kummerfeld, S.K., Teichmann, S.A. and Luscombe, N.M. 2009, A census of human transcription factors: function, expression and evolution, *Nat. Rev. Genet.*, **10**, 252–63.
 28. Ososki, A.L. and Kennelly, E.J. 2003, Phytoestrogens: a review of the present state of research, *Phytotherapy Res.*, **17**, 845–69.
 29. Shinozaki, K. 2007, Acceleration of soybean genomics using large collections of DNA markers for gene discovery, *DNA Res.*, **14**, 235.
 30. Sakai, T. and Kogiso, M. 2008, Soyisoflavones and immunity, *J. Med. Invest.*, **55**, 167–73.
 31. Tran, L.-S.P. and Nguyen, H.T. 2009, Future Biotechnology of Legumes, In: Emerich, W.D. and Krishnan, H. (eds.), *Nitrogen Fixation in Crop Production*, The American Society of Agronomy, Crop Science Society of America and Soil Science Society of America. Madison, WI, USA, pp. 265–308.
 32. Graham, P.H. and Vance, C.P. 2003, Legumes: importance and constraints to greater use, *Plant Physiol.*, **131**, 872–7.
 33. Sammut, S.J., Finn, R.D. and Bateman, A. 2008, Pfam 10 years on: 10,000 families and still growing, *Brief Bioinform.*, **9**, 210–9.
 34. Yanhui, C., Xiaoyuan, Y., Kun, H., et al. 2006, The MYB transcription factor superfamily of *Arabidopsis*: expression analysis and phylogenetic comparison with the rice MYB family, *Plant Mol. Biol.*, **60**, 107–24.
 35. Ouyang, S., Thibaud-Nissen, F., Childs, K.L., Zhu, W. and Buell, C.R. 2009, Plant genome annotation methods, *Methods Mol. Biol.*, **513**, 263–82.
 36. Altschul, S.F., Madden, T.L., Schaffer, A.A., et al. 1997, Gapped BLAST and PSI-BLAST: a new generation of protein database search programs, *Nucleic Acids Res.*, **25**, 3389–402.
 37. Umezawa, T., Sakurai, T., Totoki, Y., et al. 2008, Sequencing and analysis of approximately 40000 soybean cDNA clones from a full-length-enriched cDNA library, *DNA Res.*, **15**, 333–46.
 38. Davuluri, R.V., Sun, H., Palaniswamy, S.K., et al. 2003, AGRIS: *Arabidopsis* gene regulatory information server, an information resource of *Arabidopsis* cis-regulatory elements and transcription factors, *BMC Bioinformatics*, **4**, 25.
 39. Guo, A., He, K., Liu, D., et al. 2005, DATF: a database of *Arabidopsis* transcription factors, *Bioinformatics*, **21**, 2568–9.
 40. Gao, G., Zhong, Y., Guo, A., et al. 2006, DRTF: a database of rice transcription factors, *Bioinformatics*, **22**, 1286–7.
 41. Palaniswamy, S.K., James, S., Sun, H., Lamb, R.S., Davuluri, R.V. and Grotewold, E. 2006, AGRIS and AtRegNet: a platform to link cis-regulatory elements and transcription factors into regulatory networks, *Plant Physiol.*, **140**, 818–29.
 42. Yilmaz, A., Nishiyama, M.Y. Jr, Fuentes, B.G., et al. 2009, GRASSIUS: a platform for comparative regulatory genomics across the grasses, *Plant Physiol.*, **149**, 171–80.
 43. Li, W. and Godzik, A. 2006, Cd-hit: a fast program for clustering and comparing large sets of protein or nucleotide sequences, *Bioinformatics*, **22**, 1658–9.
 44. Higo, K., Ugawa, Y., Iwamoto, M. and Korenaga, T. 1999, Plant cis-acting regulatory DNA elements (PLACE) database, *Nucleic Acids Res.*, **27**, 297–300.
 45. Yamaguchi-Shinozaki, K. and Shinozaki, K. 2005, Organization of cis-acting regulatory elements in osmotic- and cold-stress-responsive promoters, *Trends Plant Sci.*, **10**, 88–94.
 46. Nemhauser, J.L., Mockler, T.C. and Chory, J. 2004, Interdependency of brassinosteroid and auxin signaling in *Arabidopsis*, *PLoS Biol.*, **2**, E258.
 47. Donlin, M.J. 2007, Using the generic genome browser (GBrowse), *Curr. Protoc. Bioinform.*, Chapter 9, Unit 9.9.
 48. Valliyodan, B. and Nguyen, H.T. 2006, Understanding regulatory networks and engineering for enhanced drought tolerance in plants, *Curr. Opin. Plant Biol.*, **9**, 189–95.
 49. Nakashima, K., Ito, Y. and Yamaguchi-Shinozaki, K. 2009, Transcriptional regulatory networks in response to abiotic stresses in *Arabidopsis* and grasses, *Plant Physiol.*, **149**, 88–95.
 50. Lazarova, G., Zeng, Y. and Kermod, A.R. 2002, Cloning and expression of an ABSCISIC ACID-INSENSITIVE 3 (ABI3) gene homologue of yellow-cedar (*Chamaecyparis nootkatensis*), *J. Exp. Bot.*, **53**, 1219–21.
 51. Luscombe, N.M. and Thornton, J.M. 2002, Protein-DNA interactions: amino acid conservation and the effects of mutations on binding specificity, *J. Mol. Biol.*, **320**, 991–1009.
 52. Itzkovitz, S., Tlusty, T. and Alon, U. 2006, Coding limits on the number of transcription factors, *BMC Genomics*, **7**, 239.
 53. Schlueter, J.A., Lin, J.Y., Schlueter, S.D., et al. 2007, Gene duplication and paleopolyploidy in soybean and the implications for whole genome sequencing, *BMC Genomics*, **8**, 330.
 54. Zhang, G., Chen, M., Chen, X., et al. 2008, Phylogeny, gene structures, and expression patterns of the ERF gene family in soybean (*Glycine max* L.), *J. Exp. Bot.*, **59**, 4095–107.

55. Vandepoele, K., Quimbaya, M., Casneuf, T., De Veylder, L. and Van de Peer, Y. 2009, Unraveling transcriptional control in *Arabidopsis* using *cis*-regulatory elements and coexpression networks, *Plant Physiol.*, **50**, 535–46.
56. Walther, D., Brunnemann, R. and Selbig, J. 2007, The regulatory code for transcriptional response diversity and its relation to genome structural properties in *A. thaliana*, *PLoS Genet.*, **3**, e11.
57. Zhang, W., Ruan, J., Ho, T.H., You, Y., Yu, T. and Quatrano, R.S. 2005, *Cis*-regulatory element based targeted gene finding: genome-wide identification of abscisic acid- and abiotic stress-responsive genes in, *Arabidopsis thaliana*. *Bioinformatics*, **21**, 3074–81.
58. Kim, D.W., Lee, S.H., Choi, S.B., et al. 2006, Functional conservation of a root hair cell-specific *cis*-element in angiosperms with different root hair distribution patterns, *Plant Cell*, **18**, 2958–70.
59. Won, S.K., Lee, Y.J., Lee, H.Y., Heo, Y.K., Cho, M. and Cho, H.T. 2009, *Cis*-element- and transcriptome-based screening of root hair-specific genes and their functional characterization in *Arabidopsis*, *Plant Physiol.*, **150**, 1459–73.